# SECTION-I MICROECONOMIC THEORY

CHAPTER

1

# DEMAND

## Learning Objectives

***The students will learn:***

- *The meaning and types of demand.*
- *The factors affecting demand and market demand.*
- *The law of demand, its assumptions and exceptions.*
- *The reason for the movement along the demand curve and the shift in the demand curve.*

**Course of Action**
Maximum Time = 3:15 Hrs
Maximum Questions = 85


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Movements and Shifts in Demand Curve **Page No. 11**


## Topic-1

### Demand

**Concepts covered:** Introduction to demand, factors affecting demand, demand function, Types of demand, Demand schedule and its types, Demand curve and its types, Determinants of individual and market demand

## Revision Notes

- **Demand:** The quantity of a commodity that a consumer is willing and able to buy at each possible price during a given period of time.
- **Factors Affecting Demand:**

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**Factors affecting demand**

**(1) Price of Product:** It is the most important factor affecting demand for the given commodity. There is an inverse relationship between price and quantity demanded. It means, as price increases, quantity demanded falls and vice- versa.

For example, when the price of apples reduces from ₹ 60 per kg to ₹ 40 per kg, the quantity demanded of apples increases from 1 kg to 2 kg.

**(2) Income of the Consumer:** Demand for a commodity is also affected by income of the consumer. However, the effect of change in income on demand depends on the nature of the commodity under consideration.

**(3) Tastes and Preferences:** Tastes and preferences of the consumer is directly influenced by the demand for a commodity. It include changes in fashion, customs, habits, etc. If a commodity is preferred by the consumers, then demand for such a commodity rises. On the other hand, demand for a commodity falls, if the consumers have no taste for that commodity.

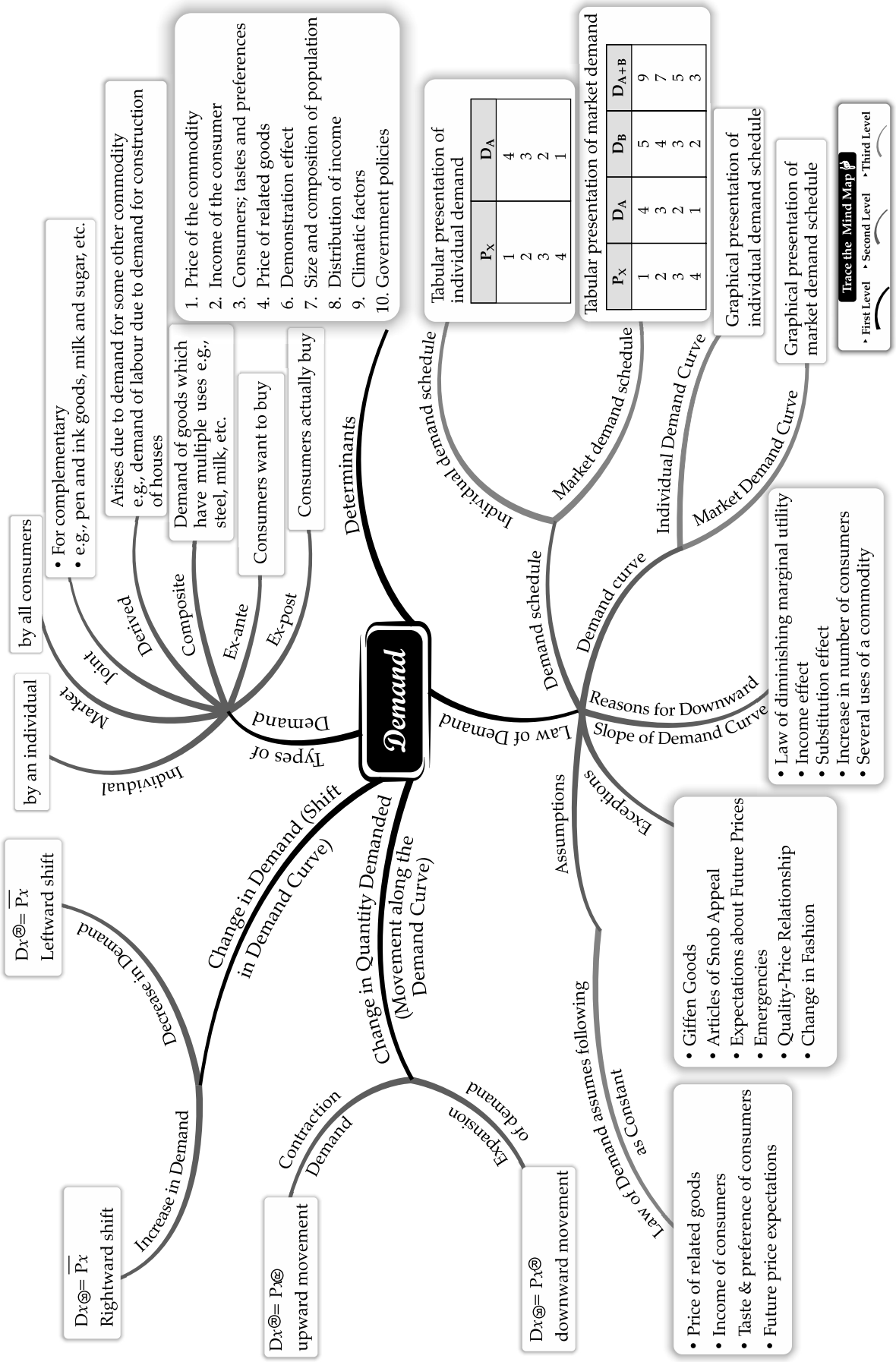
**(4) Expectation of Change in the Price in Future:** If the price of a certain commodity is expected to increase in near future, then people will buy more of that commodity. There exists a direct relationship between expectation of change in the prices in future and change in demand in the current period.

**(5) Price of Related Goods:** Demand for the given commodity is also affected by change in prices of the related goods. Related goods are of two types:

**(i)** Substitute goods

**(ii)** Complementary goods

- **Quantity Demanded:** Quantity demanded refers to a specific quantity to be purchased against a specific price of a commodity.
- **Demand Function:** Demand function represents the relationship between the quantity demanded for a




| $P_X$ | $D_A$ |
|---|---|
| 1 | 4 |
| 2 | 3 |
| 3 | 2 |
| 4 | 1 |

| $P_X$ | $D_A$ | $D_B$ | $D_{A+B}$ |
|---|---|---|---|
| 1 | 4 | 5 | 9 |
| 2 | 3 | 4 | 7 |
| 3 | 2 | 3 | 5 |
| 4 | 1 | 2 | 3 |

commodity (dependent variable) and the price of the commodity (independent variable).

- **Individual Demand Function:** Individual demand for a commodity refers to the quantity of the commodity that an individual buyer is willing to buy at a given price during a given period of time.
- **Market Demand Function:** Market demand for a good means the total quantity of a commodity that all the buyers of the good are willing to buy at a given price over a given time period.
- **Determinants of Individual Demand Function:** Important determinants of individual demand are: (1) Own price of commodity, (2) Price of related commodities, (3) Income of the consumers, (4) Tastes and preferences of consumers, (5) Expectations of consumers, etc.
- **Determinants of Market Demand Function:** Important determinants of market demand are: (1) Own price of commodity, (2) Price of related commodities, (3) Money income of the consumers, (4) Tastes and preferences of consumers, (5) Changes in weather conditions, (6) Changes in population, (7) Distribution of income, (8) Changes in structure of population, (9) Changes in quantity of money, (10) Distribution of National Wealth, (11) Phases of business cycles, (12) Change in saving habits, (13) Changes in government policies, etc.
- **Types of Demand:**
  - **Joint Demand:** The demand for two or more than two commodities which are demanded together or jointly by the consumers. For example, demand for bread and butter, car and petrol, etc. is joint demands.
  - **Derived Demand:** The demand for commodity that arises due to demand of another commodity. For example, demand for commodities like thread, buttons, etc. have derived demand that arises due to demand for clothes.
  - **Composite Demand:** The demand for a commodity that has multiple uses. For example, demand for milk, petrol, steel, plastic, etc. is composite demand.
  - **Ex-ante Demand:** The quantity of goods that a consumer 'wants to buy' during a particular period.
  - **Ex-post Demand:** The quantity of goods that a consumer 'actually buys' during a particular period.
  - **Price Demand:** It expresses the inverse functional relationship between the price and demand of a commodity, other things being equal. It is expressed as:

$$D_x = f(P_x)$$

- **Income Demand:** It expresses the direct relationship between income of the consumer and quantity demanded of a commodity, other things remaining constant. It is expressed as:

$$D_x = f(Y)$$

- **Cross Demand:** Other things being equal, when there is a change in the price of commodity X the demand for commodity Y changes, when X and Y are related goods. It is called Cross Demand. It is expressed as:

$$D_y = f(P_x)$$

- **Demand Schedule:** Demand schedule is a tabular statement that shows different quantities of a commodity that would be purchased at different prices during a given time period.
- **Individual Demand Schedule:** It represents quantity of goods demanded by an individual at different prices in a different time period.
- **Market Demand Schedule:** It is a table that shows different quantities of a commodity that all the consumers in the market are willing to buy at different possible prices of the commodity at a point of a time.
- **Demand Curve:** Demand Curve is a graphic presentation of a demand schedule showing the relationship between different quantities of a commodity demanded at different possible prices during a given period of time.
- **Individual Demand Curve:** It is a curve showing different quantities of a commodity that one particular individual buyer is ready to buy at different possible prices of a commodity at a point of time.
- **Market Demand Curve:** It is a curve showing different quantities of a commodity that all the buyers in the market are ready to buy at different possible prices of a commodity at a point of time.
- **Normal Goods:** These are those goods whose income effect is positive and price effect is negative.
- **Inferior Goods:** These are those goods whose income effect is negative.
- **Substitute Goods:** These are those goods which can be used in place of one another to satisfy human wants. For example: Tea and Coffee are substitutes. When price of a goods increases, the demand of its substitute good also increases and vice versa.
- **Complementary Goods:** These are those goods which are used together to satisfy a particular want. They complete the demand for each other. For example: Car and Petrol. There exists an inverse relationship between price and quantity demanded of complementary goods.

# SUBJECTIVE TYPE QUESTIONS

## 1 Mark Questions

**Q.1. The demand for a good increase when the income of the consumer increases. Give reason.**

**[ISC 2017]**

**Ans.** The demand for normal good increases with the increase in income as the consumer has more income to buy the good. **[Marking Scheme, ISC, 2017]**

### Examiner's Comment

- Several candidates did not mention 'real income' or 'normal goods' when income of the consumer increases.

## Answering Tip

- Determinants of demand should be clearly understood. The difference between normal goods and inferior goods must be clear.

**Q.2. Define demand? [OEB]**

**Q.3. Give the meaning of substitute goods? [OEB]**

**Ans.** Substitute goods can be referred to as those goods which are used in place of each other. For instance, tea and coffee, ball pen and gel pen.

**Q.4. Heena's salary increased, which led to her decreasing the quantity she bought of normal wheat flour and started to purchase whole grain flour. What type of good is wheat flour in this case? [OEB]**

**Ans.** As due to the increase in the salary, the demand is decreasing, in this case the wheat flour is an inferior good.

**Q.5. Saroj bought 5 kg of sugar more than usual, as it was expected that the price of sugar will rise in the near future. Which factor affecting the demand is shown in the above example? [OEB]**

**Ans.** Future Price Expectation.

**Q.6. State whether the given statement is true or false. Also give the reason for the answer. [OEB]**

**(a) "The slope of demand curve is upward sloping."**

**Ans.** False

**Reason:** The demand curve is downward sloping because with the rise in price of a goods, demand falls and with the fall in price of a goods, demand rises.

**(b) "Market demand is summation of individual demand curves."**

**Ans.** True

**Reason:** As market demand which is sum of demand by all the consumers. Hence, market demand is summation of individual demand curves.

**(c) An increase in the price of Coke would result in decrease in the demand for Pepsi.**

**Ans.** False

**Reason:** Coke and Pepsi are substitute goods. An increase in the price of Coke would induce consumers to substitute Coke by Pepsi. Demand for Pepsi will increase.

**(d) An increase in the price of sugar would result in an increase in the demand for tea.**

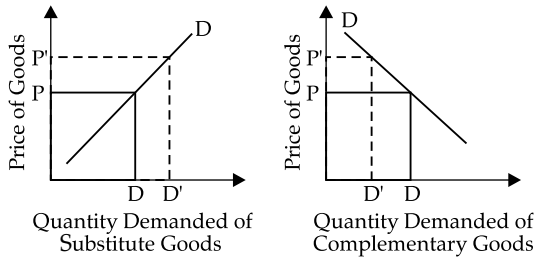
**Ans.** False

**Reason:** Sugar and tea are complementary goods. An increase in price of sugar would make tea costlier. Hence, the demand for tea would decrease.

## 2 Marks Questions

**Q.1 Draw one graph each for substitute goods and complementary goods to show the relationship between price and demand for a good. [ISC Specimen Paper 2022-23]**

**Ans.** Diagrammatic representation of both substitute goods and complementary goods




***Relationship between price and demand for a substitute good*** **-** Price of one substitute goods has positive relationship with quantity demanded of another substitute good which is represented in the above diagram.

***Relationship between price and demand for a Complementary good*** **-** Price of a complementary goods has negative relationship with quantity demanded of another complementary good.

**Q.2 With the help of a diagram explain the effect of price on the quantity demanded. [OEB]**

**Q.3 Give difference between market demand and individual demand? [OEB]**

**Ans.** Market demand can be defined as the summation of individual demand and individual demand can be defined as demand made by the individual consumer.

**Q.4 Explain the meaning of *income effect.* [OEB]**

**Q.5 Give difference between demand schedule and demand curve? [OEB]**

**Ans.** Demand schedule can be defined as the tabular representation of different quantities demanded at given prices during a given period of time. Demand curve is the graphical representation of demand schedule.

**Q.6 In the newspaper following headline appeared:**

**"Demand for goods used by old age people increased in the market."**

**Analyse the above statement with regard to the market demand? [OEB]**

**Ans.** The above statement states that the market demand has changed because of a change in the structure of the population. Due to more number of old people, the demand for goods used by them has increased.

## 3 Marks Questions

**Q.1. Explain any two factors affecting the demand of a commodity, other than its price. [ISC 2020]**

**Ans. The factors affecting the demand of a commodity, other than its price are:**

*This question is for practice and its solution is available at the end of the chapter*

(i) **Price of Related Goods:** Demand for the given commodity is also affected by change in prices of the related goods.

**Related goods are of two types:**

(a) **Substitute Goods:** Substitute goods are those goods which can be used in place of one another for satisfaction of a particular want, like tea and coffee. An increase in the price of substitute leads to an increase in the demand for given commodity and vice-versa. For example, if price of a substitute good (say, coffee) increases, then demand for given commodity (say, tea) will rise as tea will become relatively cheaper in comparison to coffee. So, demand for a given commodity is directly affected by change in price of substitute goods.

(b) **Complementary Goods:** Complementary goods are those goods which are used together to satisfy a particular want, like tea and sugar. An increase in the price of complementary good leads to a decrease in the demand for given commodity and vice-versa. For example, if price of a complementary good (say, sugar) increases, then demand for given commodity (say, tea) will fall as it will be relatively costlier to use both the goods together. So, demand for a given commodity is inversely affected by change in price of complementary goods.

(ii) **Income of the Consumer:** Demand for a commodity is also affected by income of the consumer. However, the effect of change in income on demand depends on the nature of the commodity under consideration.

(a) If the given commodity is a normal good, then an increase in income leads to rise in its demand, while a decrease in income reduces the demand.

(b) If the given commodity is an inferior good, then an increase in income reduces the demand, while a decrease in income leads to rise in demand.

(iii) **Tastes and Preferences:** Tastes and preferences of the consumer directly influence the demand for a commodity. They include changes in fashion, customs, habits, etc. If a commodity is in fashion or is preferred by the consumers, then demand for such a commodity rises. On the other hand, demand for a commodity falls, if the consumers have no taste for that commodity.

(iv) **Expectation of Change in the Price in Future:** If the price of a certain commodity is expected to increase in near future, then people will buy more of that commodity than what they normally buy. There exists a direct relationship between expectation of change in the prices in future and change in demand in the current period. For example, if the price of petrol is expected to rise in future, its present demand will increase. **[Any Two]**

## Examiner's Comments

- Common errors made by most of the candidates in attempting this part were:
  - Writing about own price of the commodity as a determinant which was irrelevant as per the question.
  - Not mentioning all the sub-cases, i.e., effect of change in income on demand for inferior and inexpensive necessary goods while writing about income and the prices of the related goods as determinants. Effect of prices of both substitute goods and complementary goods on the demand for a particular commodity was not explained (in case of prices of related goods as determinant).

## Answering Tip

- Income of the consumer as a factor affecting demand should be understood in terms of normal goods, inferior goods and inexpensive commodities. The price of related goods should be understood in terms of substitute and complementary goods.

**Q.2. How does an increase in income affect the demand for the following? [ISC 2019]**

**(i) A normal good**

**(ii) An inferior good**

**Q.3. Discuss how prices of related goods affect the demand for a commodity. [ISC 2018]**

**Ans.** Demand for the given commodity is also affected by change in prices of the related goods.

**Related goods are of two types:**

(i) **Substitute Goods:** Substitute goods are those goods which can be used in place of one another for satisfaction of a particular want, like tea and coffee. An increase in the price of substitute leads to an increase in the demand for given commodity and vice-versa. For example, if price of a substitute good (say, coffee) increases, then demand for given commodity (say, tea) will rise as tea will become relatively cheaper in comparison to coffee. So, demand for a given commodity is directly affected by change in price of substitute goods.

(ii) **Complementary Goods:** Complementary goods are those goods which are used together to satisfy a particular want, like tea and sugar. An increase in the price of complementary good leads to a decrease in the demand for given commodity and vice-versa. For example, if price of a complementary good (say, sugar) increases, then demand for given commodity (say, tea) will fall as it will be relatively costlier to use both the goods together. So, demand for a given commodity is inversely affected by change in price of complementary goods.

## Examiner's Comment

- Many candidates, instead of mentioning price of substitute goods and explaining its impact on demand for a good wrote on how the price

*This question is for practice and its solution is available at the end of the chapter*

of complementary goods impacted the demand for a substitute good. Some of those candidates who did mention price of substitute goods, did not mention whether the concerned commodity becomes relatively cheaper or dearer.

**Answering Tip**

- The students need to clarify the concept using both complementary goods and substitute goods with examples.

**Q.4 Complete the demand schedule for commodity X.**

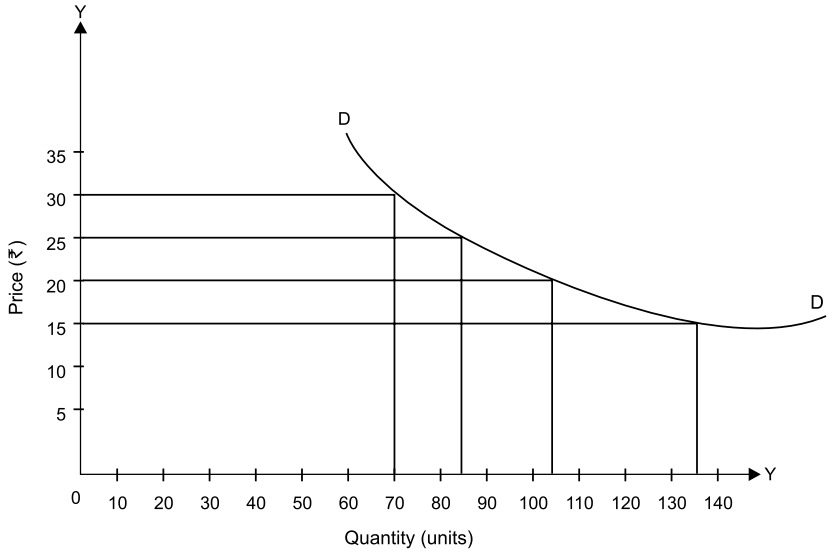
| Price (₹/unit) | Quantity demanded by Individual A (dozens) | Quantity demanded by Individual B (dozens) | Market Demand (C) (dozens) |
|---|---|---|---|
| 15 | 50 | 85 | ? |
| 20 | 45 | ? | 105 |
| 25 | ? | 45 | 85 |
| 30 | 35 | 35 | ? |

**Draw the market demand curve from the above schedule.** **[ISC 2011]**

**Ans.**

| Price (₹/unit) | Quantity demanded by Individual A (dozens) | Quantity demanded by Individual B (dozens) | Market Demand (C) (dozens) |
|---|---|---|---|
| 15 | 50 | 85 | 135 |
| 20 | 45 | 60 | 105 |
| 25 | 40 | 45 | 85 |
| 30 | 35 | 35 | 70 |

As, Quantity demanded by A + Quantity demanded by B = Market demand (C)




**Q.5. Discuss the relationship between income of the consumer and demand for a commodity with respect to normal goods, inferior goods and necessities.** **[ISC 2015]**

## 4 Marks Questions

**Q.1 Explain the concept of demand function.** **[OEB]**

**Ans.** Demand function shows the functional relationship between various determinants of demand and quantity demanded.

$D_x = f(P, Py, I, T, F)$

where,

$D_x$ = demand for commodity $x$

*This question is for practice and its solution is available at the end of the chapter*

P= Price of the commodity

$P_y$= Related good's price

I = Consumer's income

T = consumer's tastes and preferences

F= Expected change in future prices.

It shows that all the factors affecting demand have an impact on the demand for the commodity.

**Q.2 Explain the factors affecting market demand for the commodity? [OEB]**

**Ans. The factors affecting market demand for the commodity are as follows:**

**(i) Price of related commodities:** There are two types of related commodities: Substitute goods and Complementary goods.

**(a) Substitute goods:** These are those goods which can be used in place of each other. For example, Ball pen and gel pen. If there is rise in price of ball pen, then demand for gel pen will also rise and if ball pen price falls then demand for ball pen also falls.

**(b) Complementary goods:** These are those goods which are used together to satisfy a particular want. These goods are used together.

For example, Pen and Ink. If the price of pen rises, then demand for ink falls and if price of pen falls then demand for ink rises. This is because these goods are used together to satisfy a particular want.

**(ii) No. of consumers:** If there is increase in the number of consumers in the market then market demand also rises and if no. of consumers falls in the market, then market demand also falls.

**(iii) Change in structure of the population:** If in the population, no. of women rises in the market then market demand for women products also rises. And if in a population, percentage of old people is higher then demand for products used by old people will be higher.

**(iv) Distribution of income:** If in an economy, income is unequally distributed that is there is divide between rich and poor, then there will be more demand for luxuries and on the other hand, if income is equally distributed then the demand for necessities will be more.

**Q.3 Explain the different types of demand? [OEB]**

**Ans. The different types of demand are as follows:**

**(i) Price demand:** It shows that inverse relationship exists between price and quantity demanded, keeping other factors constant.

**(ii) Income demand:** It shows that other things remaining same, when income of the consumer changes then demand for the commodity also changes.

**(iii) Cross demand:** It shows that other things remaining unchanged, when price of commodity X changes then it leads to change in demand for commodity Y.

**(iv) Joint demand:** It shows that when demand for two or more commodities is jointly demanded by the consumer. **(Any Three)**

## 6 Marks Questions

**Q.1. "Market Demand depends on a lot of factors." Explain. [OEB]**

**Ans. The determinants of market demand are:**

**(i) Own price of commodity:** Price of the commodity effects the market demand in an inverse way that is, when the price of the commodity falls the demand for the commodity increases and vice versa.

**(ii) Price of related commodities:** Price of related goods, that is supplements and complements, affects the demand of the commodity.

In case of substitutes, when the price of the substitute good decreases, the quantity demanded of the commodity also decreases as it becomes relatively expensive.

In case of complements, when the price of the complement good decreases, demand of the commodity increases.

**(iii) Money income of the consumers:** When the income of the consumers influencing the market increases, so does the demand for the commodity.

**(iv) Tastes and preferences of consumers:** The fashion does rule the market. In the case of the change in the taste and preferences of the consumer, the market demand also changes according to it.

**(v) Changes in weather conditions:** If it is a sunny weather, the demand for goods required during hot weather increases in the market. In the case of rainy weather, the demand for umbrellas increases in the market. This shows that the market demand is influenced by the change in weather conditions of a place.

**(vi) Changes in population:** The demand in the market rises with the rise in the population of the place. If the population is more, then the market demand will be more and if the population is less, the market demand will be less.

**(vii) Distribution of income:** If there are more rich in the market, the demand for luxury goods will increase. If there are more poor households in the market, the demand for necessities will increase. Thus, the distribution of income in the market does influence the market demand.

**(viii) Changes in structure of population:** Change in demographics leads to a change in the market demand. If the market consists of young people, the demand for goods that are required by the youth will increase. **[Any Six]**

**Q.2. Differentiate between the following:** **[OEB]**
**(i) Joint Demand and Composite Demand**
**(ii) Ex-post Demand and Ex-ante Demand.**
**(iii) Substitute goods and Complementary goods.**

**Ans. (i) The differences between Joint Demand and Cross Demand are:**

| S. No. | Basis of Distinction | Joint Demand | Composite Demand |
|---|---|---|---|
| 1. | Meaning | The demand is said to be joint when the goods are demanded at the same time. | The demand of the commodity wherein it is put to many uses is called composite demand. |
| 2. | Examples | Tea and Sugar | Milk |

**(ii) The difference between Ex-post Demand and Ex-ante Demand are:**

| S. No. | Basis of Distinction | Ex-post Demand | Ex-ante Demand |
|---|---|---|---|
| 1. | Meaning | The quantity of goods that a consumer 'actually buys' during a particular period. | The quantity of goods that a consumer 'wants to buy' during a particular period. |
| 2. | Other name. | It is also called actual demand. | It is also called anticipated demand. |

**(iii) The differences between Substitute goods and Complementary goods are:**

| S. No. | Basis of Distinction | Substitute Goods | Complementary goods |
|---|---|---|---|
| 1. | Meaning | The goods that are used in place of one another. | The goods that are used together. |
| 2. | Examples. | Tea and Coffee | Bread and Butter |

# Topic-2 Law of Demand

**Concepts covered:** Meaning of the law of demand, Assumptions of the law of demand, Exceptions to the law of demand.

## Revision Notes

- **Law of Demand:** The law of demand states that other things remaining same, quantity demanded of a commodity is inversely related to its price. In other words, demand for a commodity rises when its price falls and its demand falls when price rises provided other factors remain unchanged.
- The law of demand states the inverse relationship between price and commodity demanded other things remaining constant (ceteris paribus).
- **Assumptions of Law of Demand:** The law of demand assumes the following as constant:
  **(i)** Price of all related goods;
  **(ii)** Income of the consumer;
  **(iii)** Tastes and preferences of consumers;
  **(iv)** Future expectations.
- **Why does Law of Demand Operate?:** (1) Law of Diminishing Marginal Utility, (2) Substitution Effect, (3) Income effect, (4) Different uses, (5) Change in number of consumers.

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Law of Demand

- **Exceptions to the Law of Demand:** (1) Giffen Goods; (2) Status Symbol Goods; (3) Fear of Shortage; (4) Fashion Related Goods; (5) Necessities of Life; (6) Conspicuous Consumption; (7) Bandwagon effect; (8) Snob effect; (9) Veblen goods.
  For example, when the price of salt increases from ₹ 20 per kg to ₹ 25 per kg, its quantity demanded remains the same because it is a necessity good.

## Key Terms

- **Giffen Goods:** Giffen goods are those inferior goods in the case of which there is a positive relationship between price and quantity demanded and income effect is positive. Law of Demand is not applicable in case of Giffen Goods.
- **Law of Diminishing Marginal Utility:** The Law of Diminishing Marginal Utility states that the additional utility gained from an increase in consumption decreases with each subsequent increase in the level of consumption.

# SUBJECTIVE TYPE QUESTIONS

## 1 Mark Questions

**Q.1. The demand curve for the consumer is downward sloping. State the reason for this feature of demand curve. [OEB]**

**Ans.** The demand curve is downward sloping because with the rise in price of a goods, demand falls and with the fall in price of a goods, demand rises.

**Q.2. When the price of oranges rise, then what happens to the demand of oranges? [OEB]**

**Ans.** The demand of oranges falls when there is rise in the price of oranges.

**Q.3. State an assumption on which the law of demand is based. [OEB]**

**Ans.** The assumptions of the law of demand is that the income of the consumer remains the same.

**Q.4. Harish said that when the price of the good decreases the relative income of the consumer increases. This will lead him to purchase more of that good. According to him, this is the reason for the downward slope of the demand curve. Which reason is he explaining for the downward slope of the demand curve? AI [OEB]**

**Ans.** Income Effect.

**Q.5. Sudha purchased gold jewellery even when the price rose in the market. Which exception to the law of demand is being shown here by Sudha? [OEB]**

**Q.6. State whether the given statements are true or false. Also, give the reason for the answer.**

**(a) "In law of demand, price of related goods and income of the consumer changes". [OEB]**

**Ans.** False

**Reason:** In law of demand, demand changes due to change in only price. But price of related goods does not change and also there is no change in income of the consumer.

**(b) "One of the exceptions of law of demand is necessity goods and luxury goods". AI [OEB]**

**Ans.** False

**Reason:** One of the exceptions of law of demand is necessity goods because with the change in price of these goods demand for them are not affected or does not change.

## 2 Marks Questions

**Q.1. What is law of demand? [OEB]**

**Ans.** The law of demand states that other things remaining same, quantity demanded of a commodity is inversely related to its price. In other words, demand for a commodity rises when its price falls and its demand falls when price rises provided other factors remain unchanged.

**Q.2. "The law of demand suffers from a lot of exception."**

**Explain any two of them. [OEB]**

**Q.3. In the newspaper following headline appeared:**

**"Due to Covid-19, demand for sanitisers, masks, etc., increased."**

**Analyse the above statement with regard to the relationship between Price and Demand. [OEB]**

**Ans.** The above situation can be described as unexpected as when the price remained same, but demand for sanitisers, masks, etc., increased because of the situation that arrived because of Covid-19.

**Q.4. There was news in the newspaper as follows:**

**"The demand for various luxury goods is high due to higher price."**

**State with reason whether the above statement follows law of demand? [OEB]**

**Ans.** The above statement does not follow the law of demand because it is the exception of law of demand.

## 3 Marks Questions

**Q.1 How does future price expectations and conspicuous consumption by the consumer act as the exception to the law of demand? [OEB]**

**Ans.** **(i) Expectations regarding future prices:** If the prices of a commodity are rising and are expected to rise further in future, the consumers tend to buy and store such commodities. As a result, demand for such commodities increases even when their price is increasing.

**(ii) Conspicuous consumption by a consumer:** The commodities like diamonds, gold, luxury cars, etc., which are considered as status symbol are not affected by the law of demand. On the other hand, they operate in reverse direction. With an increase in price of such commodities, their status value also rises and as a result the demand for such commodities rises with an increase in prices.

**Q.2 Ram wants to know how Income effect and substitution effect are the reason for the downward slope of the demand curve. Explain it to him. [OEB]**

**Ans.** **Income Effect**: It refers to change in quantity demanded of a commodity when real income of the consumer changes owing to change in own price of the commodity. When the price of a commodity falls, the real income of the consumer increases which can be used to buy more quantity of the commodity.

*This question is for practice and its solution is available at the end of the chapter*

**Substitution Effect**: It refers to change in quantity demanded of commodity-X when relative price of the commodity $\left(\frac{P_z}{P_y}\right)$ changes owing to change in $P_x$. When the price of a commodity falls, the consumers of the substitute goods switch over to the commodity whose price has fallen. As a result, demand of the commodity increases.

Therefore, both income effect and substitution effect are the reasons for the downward slope of the demand curve.

**Q.3. Explain the law of demand with the help of schedule? [OEB]**

## 6 Marks Questions

**Q.1. Explain any four exceptions to the law of demand. [OEB]**

**Ans. (i) Giffen Goods:** These are inferior goods on which the consumer spends a large part of his income and their demand rises with an increase in price and demand falls with decrease in price. For example, in our country, it is often seen that when price of coarse cereals like jowar and bajra falls, the consumers have a tendency to spend less on them and shift over to superior cereals like wheat and rice. This phenomenon, popularly known as' Giffen's Paradox' was first observed by Sir Robert Giffen. A Giffen good is a good for which demand increases as the price increases and falls when the price decreases. A Giffen good has an upward-sloping demand curve. A Giffen good is typically an inferior product that does not have easily available substitutes, as a result of which the income effect dominates the substitution effect.

**(ii) Status Symbol Goods:** The exception relates to certain prestige goods which are used as status symbols. For example, diamonds, gold, antique paintings, etc., are bought due to the prestige they confer upon the possessor. These are wanted by the rich persons for prestige and distinction. The higher the price, the higher will be the demand for such goods.

**(iii) Fear of Shortage:** If the consumers expect a shortage or scarcity of a particular commodity in the near future, then they would start buying more and more of that commodity in the current period even if their prices are rising. The consumers demand more due to fear of further rise in prices. For example, during emergencies like war, famines, etc., consumers demand goods even at higher prices due to fear of shortage and general insecurity.

**(iv) Fashion Related Goods:** Goods related to fashion do not follow the law of demand and their demand increases even with a rise in their prices. For example, if any particular type of dress is in fashion, then demand for such dress will increase even if its price is rising.

**(v) Necessities of Life:** Another exception occurs in the use of such commodities, which become necessities of life due to their constant use. For example, commodities like rice, wheat, salt, medicines, etc., are purchased even if their prices increase.

**(vi) Conspicuous Consumption:** This exception to the law of demand is associated with the doctrine propounded by Thorsten Veblen. A few goods like diamonds, etc., are purchased by the rich and wealthy sections of the society. The higher the price of the diamond the higher the prestige value of it. So when price of these goods falls, the consumers think that the prestige value of these goods comes down. So quantity demanded of these goods falls with fall in their price. So the law of demand does not hold good here.

**(vii) Bandwagon Effect:** Bandwagon effect refers to the desire or demand for a good by a person who wants to be in style because possession of a good is in fashion and therefore many others have it. Thus, in a bandwagon effect, the quantity demanded of a good that an individual buys increases in response to the increase in the quantity purchased by other individuals.

**(viii) Snob Effect:** Snob effect refers to the desire to possess a unique commodity having a prestige value. The snob effect is described as a situation where the demand for a certain good by individuals of a higher income level is inversely related to its demand by those of a lower income level. The demand curve will have a positive slope, rather than the typical negatively sloped demand curve of normal goods. This situation is derived by the desire to own unusual, expensive or unique goods. These goods usually have a high economic value, but low practical value. The less of an item available, the higher its snob value. Examples of the snob effect are rare works of art, especially designed sport cars, especially designed clothing made to order, very expensive luxury cars.

**(ix) Veblen Goods:** These are types of luxury goods for which the quantity demanded increases as the price increases, an obvious opposition of the law of demand. Consumers actually prefer more of the good as its price rises, and the result is an upward sloping demand curve. **[Any Four]**

**Q.2. Explain any four reasons for the demand curve to be downward sloping. [ISC 2017]**

**Ans. The reasons for the demand curve to be downward sloping are:**

**(i) Law of diminishing marginal utility:** A consumer always equalises marginal utility with price. The law states that a consumer derives less and less satisfaction (utility) from every additional increase in the stock of a commodity. When price of a commodity falls the consumer's price utility equilibrium is disturbed i.e., price becomes smaller than utility. The consumer in order to restore the

*This question is for practice and its solution is available at the end of the chapter*

new equilibrium between price and utility buys more of it so that, the marginal utility falls with the rise in the amount demanded. So long the price of a commodity falls, the consumer will go on buying more amount of it so as to reduce the marginal utility and make it equal with new price. Thus, the shape and slope of a demand curve is derived from the slope of marginal utility curve.

(ii) **Income effect:** Another cause behind the operation of law of demand is income effect. As the price of a commodity falls, the consumer has to buy the same amount of the commodity at less amount of money. After buying his required quantity he is left with some amount of money. This constitutes his rise in his real income. This rise in real income is known as income effect. This increase in real income induces the consumer to buy more of that commodity. Thus, income effect is one of the reasons why a consumer buys more at falling prices.

(iii) **Substitution effect:** When the price of a commodity falls, it becomes relatively cheaper than other commodities. The consumer substitutes the commodity whose price has fallen for other commodities which becomes relatively dearer. For example with the fall in price of tea, coffee's price being constant, tea will be substituted for coffee. Therefore, the demand for tea will go up.

(iv) **Size of consumer group:** When the price of a commodity falls many other consumers who were deprived of that commodity at the previous price become able to buy it now as the price comes within their reach. For example the units of colour TV. increases with a remarkable fall in price of it. The opposite will happen with a rise in prices.

(v) **Different use of commodity:** There are some commodities which have multiple uses. Their uses depend upon their respective prices. When their prices rise they are used only for certain selected purposes. That is why their demand goes down. For example electricity can be put to different uses like heating, lighting, cooling, cooking, etc. If its price falls people use it for other uses other than that. A rise in price of electricity will force the consumer to minimise its use. Thus, with a fall and rise in price of electricity its demand rises and falls accordingly.

**[Any Four]**

### Examiner's Comment

- Under the law of diminishing marginal utility, a number of candidates related the law but did not establish the relation between price and demand. Several candidates did not explain substitution effect correctly. In some cases, determinants of demand were explained which was not required.

### Answering Tip

- The students need to understand that without price and demand the law of diminishing marginal utility will be incomplete.

**Q.3. What is stated in the law of demand? State the assumptions to the Law of Demand. Explain any two exception to the law of the demand.**

**[OEB]**

**Ans. Law of Demand:** The law of demand states that other things remaining same, quantity demanded of a commodity is inversely related to its price. In other words, demand for a commodity rises when its price falls and its demand falls when price rises provided other factors remain unchanged.

**Assumptions of Law of Demand**: The Law of Demand assumes the following as constant:

(i) Price of all related goods

(ii) Income of the consumer

(iii) Tastes and preferences of consumers

(iv) Future expectations

The demand for normal good increases with increase in income as the consumer has more income to buy the good.

## Topic-3 Movements and Shifts in Demand Curve

**Concepts covered:** Extension and Contraction of Demand, Increase and Decrease in Demand

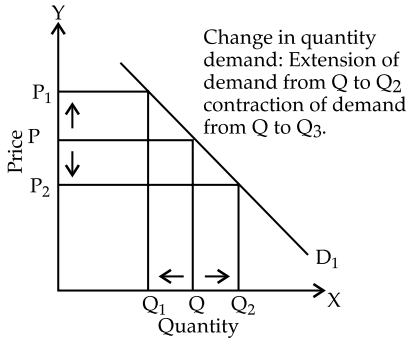
## Revision Notes

- **Movement along the Demand Curve:** It occurs when changes in quantity demanded are related to changes in price of the commodity other things remains constant.
  - When there is decrease in price of a commodity there is increase in demand of that commodity. This is called extension of demand.
  - When there is increase in price of a commodity, there is decrease in the demand for that commodity. This is called contraction of demand.
  - Thus demand varies in opposite direction due to change in price.
- **Change in Demand or Shift in Demand Curve:** It refers to increase or decrease in demand for a commodity in response to change in other determinants of demand other than the price of the commodity.

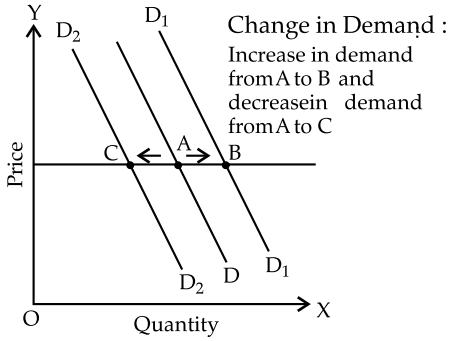
Scan to know more about this topic

Movement and Shifts in Demand Curve

- When more is demanded at the same price, there is rightward shift in demand curve and is a situation of Increase in demand. When less is demanded at the same price it is a situation of decrease in demand and there is leftward shift in demand curve.




- Let us take an example of tea and coffee. If there is a fall in the price of the coffee then the demand for tea will fall. The change (decrease) in demand for tea is due to the fall in the price of its substitute (coffee in this case).




# SUBJECTIVE TYPE QUESTIONS

## 1 Mark Questions

**Q.1. Give reason: X and Y are substitute goods. A rise in the price of X results in a rightward shift of the demand curve of Y.** AI **[ISC 2017]**

**Ans.** A rise in the price of x will result in the **fall in the demand** of x and rise in the demand of y. The demand curve of y will **shift to the right**.
**[Marking Scheme, ISC, 2017]**

**Q.2. Define movement along the demand curve?**

**Q.3. What will be the effect on the demand curve when the income of the consumer rises? [OEB]**

**Ans.** The demand curve will shift to the right.

**Q.4. A very famous actress starts wearing very different type of clothes. How will this effect the demand curve for other clothes available in the market?** AI **[OEB]**

**Ans.** The demand curve of the other clothes will **shift to the left.**

**Q.5. The price of tea rises. What will be the effect of this on the demand curve of coffee? [OEB]**

**Ans.** The demand curve of coffee will **shift to the right**.

**Q.6 State whether the given statement is true or false. Give reason for the answer.**
**" There is rise in demand due to fall in price , and it is called as increase in demand". [OEB]**

**Ans.** False
**Reason:** Rise in demand due to fall in price is called as expansion in demand and not increase in demand.

## 2 Marks Questions

**Q.1. Distinguish between increase in demand and increase in quantity demanded of a good. [OEB]**

**Ans.** Increase in Demand and Increase in Quantity Demanded

**(i)** Increase in demand refers to increase in the purchase of a commodity at its existing price. Increase in quantity demanded refers to increase in the purchase of a commodity due to a full in its price.

**(ii)** Increase in demand occurs due to change in factors other than price of the commodity. Increase in quantity demanded occurs due to change in price of the commodity.

**(iii)** Diagrammatically, an increase in demand is shown by a forward shift in demand curve while increase in quantity demanded is shown by a downward movement on the same demand curve.

**Q.2. Difference between movements along a demand curve and shift of a demand curve. [OEB]**

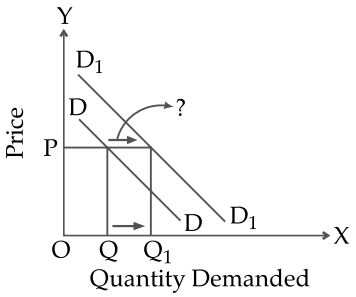
**Ans.**

| Movements along a demand curve | Shift of a demand curve |
|---|---|
| It involves only one demand curve. | It involves two demand curve. |
| It is brought about by changes in the quantity demanded. | Brought about a changes in other factors that influences demand other the price of the product. |
| A different quantity is demanded only at a different price. | A different quantity is demanded at the same price as before. |
| A movement along the curve can be traced up and down along the same curve. | A shift causes to move either to the right of left. |

**(Any Two)**

*This question is for practice and its solution is available at the end of the chapter*

**Q.3 What does the below diagram represent with explanation? [OEB]**



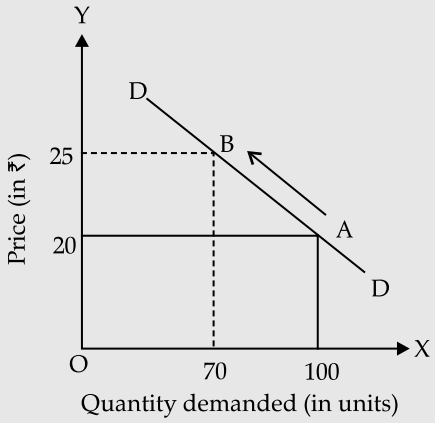
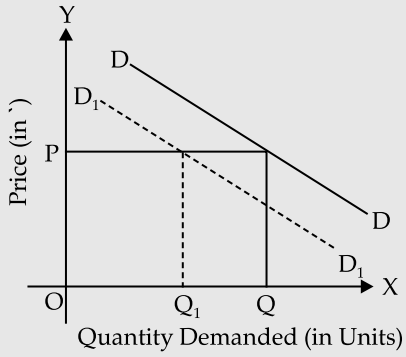

**Ans.** The above diagram represents increase in demand. Increase in demand can be defined as when demand increases due to change in other factors keeping price constant. These other factors can be a rise in price of substitute goods, a fall in price of complementary goods, an increase in income in case of normal goods, a fall in income in case of inferior goods, or favourable change in tastes and preferences of the consumer.

**Q.4. Rita was saying to Misha that "Demand of laptops increased due to change in technique of production of various services and also because people have to work from home due to covid-19". Analyse from above what concept Rita was pointing towards? [OEB]**

## 3 Marks Questions

**Q.1. Differentiate between contraction of demand and decrease in demand, using diagrams. [ISC 2018]**

**Ans. The difference between contraction of demand and decrease in demand are:**

| Contraction of demand | Decrease in demand |
|---|---|
| Other things being equal, with the rise in price, demand for a commodity falls, it is called contraction of demand. | A decrease in demand implies that at a given price a less quantity is purchased due to change in other factors, such as: |
| | A decrease in demand implies that at a given price a less quantity is purchased due to change in other factors, such as:<br>• Decrease in consumer's income.<br>• Decrease in population.<br>• When price of the commodity is expected to fall in near future.<br>• Rise in the price of complementary goods.<br>• Fall in price of substitute goods<br>• Unfavourable change in taste and preference. (any one of the factors) |
| Contraction in Demand   | Decrease in Demand   |

**[Marking Scheme, ISC, 2018]**

### Examiner's Comment

- Several candidates, instead of writing that contraction of demand means fall in demand / decrease in demand due to rise in price, wrote that fall in demand / decrease in demand is due to change in price. Many candidates either drew the supply curve or showed the extension in demand.

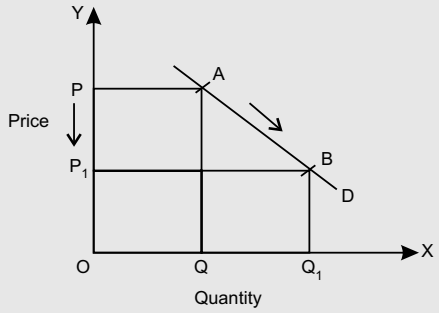
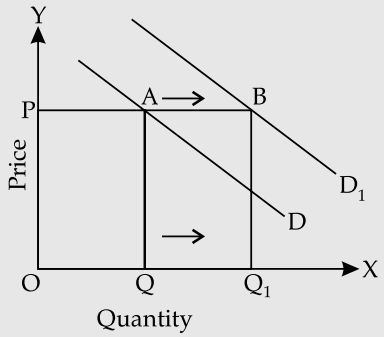
*This question is for practice and its solution is available at the end of the chapter*

## Answering Tips

- Understand clearly the concepts of contraction of demand and decrease in demand.
- Students need to practice drawing diagrams.

**Q.2. Differentiate between extension of demand and increase in demand, using diagrams. [ISC 2015]**

**Ans.**

| Extension of demand | Increase in demand |
|---|---|
| Extension of demand is due to fall in price (own price) | Increase of demand is due to other factors like income increase, favourable tastes, etc. |
|   |   |

**[ISC Marking Scheme 2015]**

## Examiner's Comment

- A large number of candidates answered this part well but some did not show the movement and shift with the help of arrows.

## Answering Tip

- Practice more for understanding the concepts of extension of demand and increase in demand. The diagrams should be properly labelled and arrows should be used to show movement/ shift.

**Q.3. With the help of diagrams, show the shift and movement on the demand curve. [OEB]**

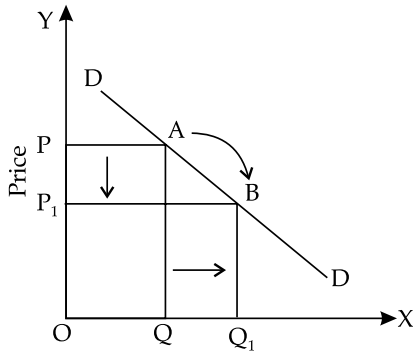
**Q.4. Explain the concepts of contraction in demand and decrease in demand. [OEB]**

**Ans.** Contraction in demand is based on Law of Demand which states that quantity demanded for the commodity falls due to the rise in price of the commodity. The fall in quantity demanded due to the rise in price of the commodity is known as contraction in demand.

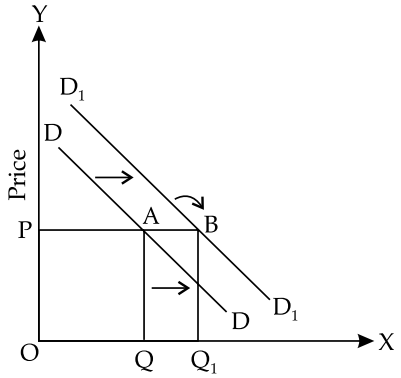
**Decrease in demand:** A decrease in demand means that consumers now demand less at a given price of a commodity. Decrease in demand is due to unfavourable change in factors other than price of the commodity. In case of decrease in demand, the demand curve shifts towards left.

## 6 Marks Questions

**Q.1. Study the figures given below and answer the questions that follows: [OEB]**




Quantity Demanded
Increase in Demand




Quantity Demanded
Extension of Demand

*This question is for practice and its solution is available at the end of the chapter*

**(a) Explain the situations as shown in both the figures.**

**Ans. Left Figure:** When the price of the commodity demanded decreases, there is an increase in the quantity demanded of that commodity. In such a situation, we call it the Extension of demand of the commodity. As shown in the figure when the price fell from P to $P_1$ the quantity demanded increased from Q to $Q_1$.

**Right Figure:** When the quantity demanded by the consumer increases not due to the rise in the price but due to the increase in other factors of demand, it leads to the rightward shift in the demand curve, which is also called the increase in demand.

**(b) Differentiate between the two situations as shown in the figures.**

**Ans. The differences are:**

| S. No. | Basis of Distinction | Increase in Demand | Expansion in Demand |
|---|---|---|---|
| **(i)** | Meaning | When there is a shift in the demand curve to the right then it means the entire demand curve has shifted to the right side. | Expansion of demand shows the changes on the same demand curve. Here there is a downward movement along the same demand curve. |
| **(ii)** | Reason | The demand curve shifts to the right when there are changes in factors other than the price of the product resulting in increase in demand. | Expansion of demand occurs when there is a fall in the price of the product. |
| **(iii)** | Effect of Substitute | Let us take an example of tea and coffee. If there is a rise in the price of the coffee then the demand for tea will rise. The change (increase) in demand for tea is due to the rise in the price of its substitute (coffee in this case). | If the price of tea is decreased than this will result in more purchases of tea. Here the only reason being the price (all the other factors are kept constant) will cause the variation (rise) in demand. |
| **(iv)** | Price Factor | The demand curve shifts to the right due to changes in the non-price factors. In this case it may be due to substitution effect. | Extension in demand is caused only due to the price factor. |

# OBJECTIVE TYPE QUESTIONS

## [A] Multiple Choice Questions

**Q.1. How would rational consumers react to a situation when the price of the car remains constant but the price of the petrol increases?**

**[ISC Specimen Paper, 2023]**

**(A)** They would tend to buy less cars.

**(B)** They would tend to buy more cars.

**(C)** They would not react to the situation.

**(D)** They would wait for the price of petrol to come down.

**Ans. Option (A) is correct.**

*Explanation:* Petrol and car are complementary goods and are used together. Thus, when the price of petrol increases the relative price of using the car increases for the consumer, even though the price of car remains constant. This leads to the fall in the demand for car that means the consumers would buy less car.

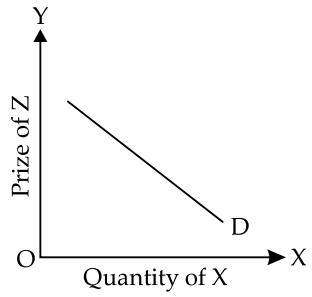
**Q.2. Identify the good that has a direct price-demand relationship.** **[ISC Specimen Paper, 2023]**

**(A)** Giffen Good

**(B)** Complementary Good

**(C)** Normal Good **(D)** Inferior Good

**Q.3. The figure given below shows the relation between the quantity demanded for the goods X and the price of the goods Z. What type of goods are X and Z?** **[ISC 2023]**




**(A)** Unrelated goods

**(B)** Substitute goods

**(C)** Complementary goods

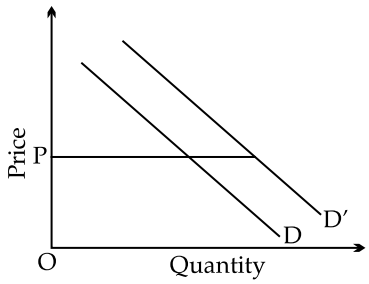
**(D)** Giffen goods

**Ans. Option (C) is correct.**

*Explanation:* The given figure shows the demand curve of complementary goods of Z. Demand for a complementary good X decreases when the price of the commodity Z rises.

*This question is for practice and its solution is available at the end of the chapter*

**Q.4.**




**In the figure given above, D is demand curve for normal good. It shifts to D′ which is a case of:** **[ISC Specimen Paper, 2022-23]**

**(A)** extension of demand due to change in price level.
**(B)** increase in demand due to rise in expected future price of the commodity.
**(C)** contraction of demand due to increase in price of the substitute.
**(D)** increase in demand due to rise in price of the complimentary good.

**Ans. Option (B) is correct.**

**Q.5. Which of the following is not an exception to the law of demand?** **[ISC Sem-I, 2022]**

**(A)** Articles of snob value. **(B)** Veblen effect
**(C)** Substitution effect. **(D)** Giffen Good.

**Ans. Option (C) is correct.**

*Explanation:* Substitution effect is the reason for the downward slope of the demand curve, thus it is not an exception to the law of demand.

**Q.6. An increase in income leads to decrease in demand for instant noodles, then instant noodles is:** **[ISC Sem-I, 2022]**

**(A)** Normal good.
**(B)** Inferior good.
**(C)** Substitute good.
**(D)** Complementary good.

**Ans. Option (B) is correct.**

*Explanation:* As with the increase in the income of the consumer, the demand for the instant noodles decreases then the instant noodles are inferior goods as inferior goods show an inverse relationship between income of the consumer and the quantity demanded.

**Q.7. When price is ₹5/- per unit of a commodity. A's demand is 11 units, B's Demand is 14 units and C's demand is 8 units, then market demand will be:** **AI [ISC Sem-I, 2022]**

**(A)** 11 units **(B)** 14 units
**(C)** 17 units **(D)** 33 units

**Ans. Option (D) is correct.**

*Explanation:*

Market Demand = Demand of A + Demand of B + Demand of C
= 11 + 14 + 8 = 33 units.

**Q.8. Which factor causes the demand for a commodity to decrease?** **[ISC Sem-I, 2022]**

**(A)** Increase in income.
**(B)** Fall in price of substitute goods.
**(C)** Expectation that prices will increase.
**(D)** Increase in market size.

**Ans. Option (B) is correct.**

*Explanation:* When there is a fall in the price of substitute goods, the good becomes relatively expensive, thus the consumer will shift the demand to the substitute good, decreasing the demand for the commodity.

**Q.9. What relation do you notice between Price and Demand in the table given below:** **[ISC Specimen Paper, Term-I, 2021-22]**

| Price | Demand (Units) |
|---|---|
| 10 | 20 |
| 8 | 25 |

**(A)** Increase in demand.
**(B)** Decrease in demand.
**(C)** Extension in demand.
**(D)** Contraction in demand.

**Ans. Option (C) is correct.**

*Explanation:* As per the table, we can see that there is an increase in the demand with the fall in the price. This means that there is a downward movement along the demand curve, implying an extension in demand.

**Q.10. Which of the given options is a reason of change in demand?** **[ISC Specimen Paper, Term-I, 2021-22]**

**(A)** Change in Consumer's Income
**(B)** Change in Prices of Related Goods
**(C)** Population increase
**(D)** All of the above

**Ans. Option (D) is correct.**

**Q.11. The reason for decrease in demand is:** **[ISC Specimen Paper, Term-I, 2021-22]**

**(A)** Rise in price
**(B)** Increase in the price of substitute goods.
**(C)** Increase in the price of complementary goods.
**(D)** Decrease in income in case of inferior goods.

**Ans. Option (C) is correct.**

**Q.12. The reason for downward sloping demand curve is:** **[ISC Specimen Paper, Term-I, 2021-22]**

**(A)** Inverse relation between price and demand
**(B)** Inverse relation between income and demand
**(C)** Direct relation between price and demand
**(D)** Both (A) and (B)

**Ans. Option (A) is correct.**

**Q.13. If the price of petrol increases, the demand for cars will:** **AI [ISC Specimen Paper, Term-I, 2021-22]**

**(A)** Increase **(B)** Decrease
**(C)** Expand **(D)** Vary

**Ans. Option (B) is correct.**

*Explanation:* Car and Petrol are complementary goods. Thus, when the price of petrol increases, the demand for car decreases as using the car becomes relatively expensive.

**Q.14. What is Ex-ante demand also known as?** **[ISC Specimen Paper, Term-I, 2021-22]**

**(A)** Intended demand
**(B)** Actual demand
**(C)** Effective demand
**(D)** Unplanned demand

**Ans. Option (A) is correct.**

*Explanation:* Ex-ante demand is the quantity of goods that a consumer 'wants to buy' during a particular period. Thus, it is the intended demand.

**Q.15. _____ is an example of goods of Snob Appeal.** **[ISC Specimen Paper, Term-I, 2021-22]**

**(A)** Buying a Ferrari
**(B)** Buying basic utilities
**(C)** Buying a car insurance
**(D)** Opening a bank account

**Ans. Option (A) is correct.**
*Explanation:* Snob Appeal is when the commodity is bought to serve as a status symbol. Buying a Ferrari, serves as a status symbol, so it is an example of snob appeal.

**Q.16. The demand for a commodity refers to:**
**[ISC Specimen Paper, Term-I, 2021-22]**
**(A)** desire for a commodity.
**(B)** need for a commodity.
**(C)** quantity demanded at the lowest prices during a particular period of time.
**(D)** quantity demanded at various possible prices during a particular period of time.

**Ans. Option (D) is correct.**

**Q.17. Giffen goods are:**
**[ISC Specimen Paper, Term-I, 2021-22]**
**(A)** Normal goods, whose demand falls with a rise in their price.
**(B)** Inferior goods, whose demand falls with a rise in real income.
**(C)** Inferior goods, whose demand falls with a fall in their price.
**(D)** Superior goods, whose demand falls with a rise in real income.

**Ans. Option (B) is correct.**
*Explanation:* Giffen Goods are special kind of inferior goods on which the consumer spends a large part of his income, and their demand rises with an increase in price and demand falls with decrease in price.

**Q.18. "The slope of demand curve is upward sloping."**
**(True or False) [OEB]**

**Ans.** False
*Explanation:* The given statement is false. The demand curve is downward sloping because with the rise in price of a goods, demand falls and with the fall in price of a goods, demand rises.

**Q.19 "Market demand is summation of individual demand curves."** **(True or False) [OEB]**

**Ans.** True
*Explanation:* The given statement is true. As Market demand which is sum of demand by all the consumers. Hence, market demand is summation of individual demand curves.

**Q.20. An increase in the price of Coke would result in decrease in the demand for Pepsi.**
**(True or False) [OEB]**

**Ans.** False
*Explanation:* Coke and Pepsi are substitute goods. An increase in the price of Coke would induce consumers to substitute Coke by Pepsi. Demand for Pepsi will increase.

**Q.21. An increase in the price of sugar would result in an increase in the demand for tea.**
**(True or False) [OEB]**

**Ans.** False
*Explanation:* Sugar and tea are complementary goods. An increase in price of sugar would make tea costlier. Hence, the demand for tea would decrease.

**Q.22. An increase in the income of a consumer would result in an increase in demand for all types of goods that are demanded by a consumer.**
**(True or False) [OEB]**

**Ans.** False
*Explanation:* With the increase in income, the consumer's demand for normal goods will increase. Demand for inferior goods may, in fact, fall.

**Q.23. "In law of demand, price of related goods and income of the consumer changes".**
**(True or False) [OEB]**

**Ans.** False
*Explanation:* In law of demand, demand changes due to change in only price. All other factors affecting demand are taken to be constant.

**Q.24. "One of the exception of law of demand is necessity goods".** **(True or False) [OEB]**

**Ans.** True
*Explanation:* One of the exception of law of demand is necessity goods because with the change in price of these goods demand for them are not affected or does not change.

## [B] Assertion Reason Questions

**Directions: In the following questions, a statement of Assertion (A) is followed by a statement of Reason (R). Mark the correct choice as:**
**(A)** Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
**(B)** Both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A).
**(C)** Assertion (A) is true but Reason (R) is false.
**(D)** Assertion (A) is false but Reason (R) is true.

**Q.1. Assertion (A):** If due to fall in the price of goods X, demand for goods Y rises, the two goods are complementary.
**Reason (R):** Complementary goods have inverse relation. Alternatives: **[OEB]**

**Ans. Option (C) is correct.**
*Explanation:* When due to the fall in the price of a good leads to the rise in the demand of the related goods they are complements. Thus complements show a positive relation with each other.

**Q.2. Assertion (A):** The demand curve slopes downward because of law of diminishing marginal utility.
**Reason (R):** More of a product is consumed the marginal (additional) benefit to the consumer rises. **[OEB]**

**Q.3. Assertion (A):** A goods is considered normal when with rise/fall in income of its consumer, its demand rises/falls.
**Reason (R):** Normal goods have negative income effect. **[OEB]**

**Ans. Option (C) is correct.**
*Explanation:* Normal goods have positive income effect as when the income of the consumer increases the demand for the normal good increases.

**Q.4. Assertion (A):** The demand for tea increased when the price of coffee increased.
**Reason (R):** Tea and coffee are substitute goods. **[OEB]**

**Ans. Option (A) is correct.**
*Explanation:* Substitute goods are those goods that are used in place of one another. When the price of tea increases, the demand for coffee increases and vice versa, as tea and coffee are substitutes.

*This question is for practice and its solution is available at the end of the chapter*

**Q.5. Assertion (A):** The fashion prevailing in the market determines the demand for clothes.

**Reason (R):** Change in taste and preference is an important determinant of demand. **[OEB]**

**Ans. Option (A) is correct.**

*Explanation:* Demand of clothes changes as per the change in fashion, as the change in fashion implies that there is a change in the taste of the consumer regarding that particular type of cloth irrespective of the price of the commodity.

# Case Based Questions

## [A] Multiple Choice Questions

**Read the following passage and answer the questions that follows:**

The Coca-Cola Company is an American multinational beverage company, with its headquarters in Atlanta, Georgia. The first company that conducted its operation in the soft drink industry was Coca-Cola. It is the world's largest non-alcoholic beverage company serving more than 1.8 billion consumers daily in more than 200 countries. It has a portfolio of more than 3,500 (more than 800 no or low calorie) products. However, the company is best known for its flagship product Coca-Cola which was originally intended to be a patented medicine was invented in 1886 by pharmacist John Smith Pemberton in Columbus, Georgia. The Coca-Cola products can be termed as normal goods and in August, 2019 Coca-Cola has introduced the new product into the market,that is, zero sugar where the demand has increased for the product in the market.

According to the council of Australian Food Technology association and Institute of Food Science and Technology, the Australian non-alcoholic beverages industry has been growing steadily, with 2.3 percent increase in overall production in the year 2000 which amounts to 2.25 billion litres. However, in the recent years, sales of customary carbonated soft drinks have dropped as more and more customers become health conscious and move away from high-calorie sugary drinks. Soft Carbonated drinks and other alcohol free beverages manufacturers have also sensed the effects of intensifying competition from private-label soft drink makers. Nevertheless, sales of greater value energy and sports drinks have driven profit generation in the industry. **[OEB]**

**(i) The demand for Coca-Cola is ..... in the present times.**

**(A)** Elastic Demand **(B)** Inelastic Demand
**(C)** Unitary Elastic **(D)** Zero Elastic.

**Ans. Option (A) is correct.**

*Explanation:* The demand for Coca-Cola is elastic because it has close substitutes available, and when price of one substitute rises, then demand for given good also rises.

**(ii) The demand for coca-cola has .....**

**(A)** increased **(B)** decreased
**(C)** remained same **(D)** can't say

**Ans. Option (B) is correct.**

**(iii) ..... is the want to buy a product backed by purchasing power.**

**(A)** Utility **(B)** Supply
**(C)** Production **(D)** Demand.

**Ans. Option (D) is correct.**

*Explanation:* Demand can be defined as the want to buy a commodity which is supported with the purchasing power. Demand is affected by various factors such as price of related goods, income of the consumers, tastes and preferences of consumer etc.

**(iv) What type of good is coca - cola?**

**(A)** Normal **(B)** Inferior
**(C)** Luxury **(D)** none of the above

**Ans. Option (A) is correct.**

*Explanation:* Coca-cola can be considered as a normal good as with increase in income of the consumer, demand for coca-cola also increases. And there is inverse relationship between price of coca-cola and demand for it.

**(v) What type of goods are Coca - cola and Pepsi?**

**(A)** Complements **(B)** Substitutes.
**(C)** Giffen **(D)** Veblen

**Ans. Option (B) is correct.**

*Explanation:* Substitute goods are those goods which can be used in place of each other. For example, coca-cola and pepsi, tea and coffee, etc. If the price of one good rises then demand for other goods also rises and when prices of one good falls then demand for other goods also falls.

## [B] Subjective Based Questions

**I.** Manoj bought 2 kg of mangoes when the price of the mangoes was ₹150 per kilo. Next time when he bought mangoes, the price went down to ₹100 per kilo and so he bought 4 kg of mangoes. Next again when the price went down to ₹75 per kilo, he bought 7 kg of mangoes.

Shrishant bought 5 L of milk. He used 1 L of milk to make some rasogulla and then 3 L of the milk to make some sweets. Remaining 1 L he used in drinking and making tea. **[OEB]**

**(a) What is the relationship shown between the price and the demand of mangoes? Does it fulfil the law of demand?**

**Ans.** There is an inverse relationship shown between the price and the demand of mangoes as shown in the extract.

Yes it fulfils the law of demand.

**(b) State the law of demand.**

**Ans.** The law of demand states that, other things remaining constant, when the price of a commodity falls, the quantity demanded of the commodity rises and when the price of the commodity rises, the quantity demanded of the commodity falls.

**(c) Identify and define the type of demand as shown by Shrishant.**

**Ans.** The type of demand as shown by Shrishant is composite demand. Composite demand is the demand of the commodity wherein it is put to many uses. As in the extract, Shrishant is using milk in numerous ways, thus it is an example of composite demand.

**(d) Which factor affecting demand is shown in the above extract? Explain.**

**Ans.** The factor as shown in the above extract is the Price of the commodity.

Price of the commodity is the most important factor affecting demand for the given commodity. Generally, there exists an inverse relationship between price and quantity demanded. It means, as price increases, quantity demanded falls due to decrease in the satisfaction level of consumers.

**(e) Explain any other factor affecting demand not shown in the above extract.**

**Ans. The factors affecting market demand for the commodity are as follows:**

**(i) Price of related commodities:** There are two types of related commodities: Substitute goods and Complementary goods.

**(a) Substitute goods:** These are those goods which can be used in place of each other. For example, Ball pen and gel pen. If there is rise in price of ball pen, then demand for gel pen will also rise and if ball pen price falls then demand for ball pen also falls.

**(b) Complementary goods:** These are those goods which are used together to satisfy a particular want. These goods are used together.
For example, Pen and Ink. If the price of pen rises, then demand for ink falls and if price of pen falls then demand for ink rises. This is because these goods are used together to satisfy a particular want.

**(ii) No. of consumers:** If there is increase in the number of consumers in the market then market demand also rises and if no. of consumers falls in the market, then market demand also falls.

**(iii) Change in structure of the population:** If in the population, no. of women rises in the market then market demand for women products also rises. And if in a population, percentage of old people is higher then demand for products used by old people will be higher.

**(iv) Distribution of income:** If in an economy, income is unequally distributed, that is, there is divide between rich and poor, then there will be more demand for luxuries and on the other hand, if income is equally distributed then the demand for necessities will be more.

**II.** When we talk about the law of demand, it does not follow always. Like if we see the gold prices increasing every day and still a lot of people instead of decreasing the consumption, they rather increase the demand and buy more gold. Not only that, during COVID it was seen that the people were hoarding a lot of food items and other essentials at home so that they don't face the shortage. Likewise there are other ways in which the law of demand is challenged.
It is not that the law of demand is just there for no reason, it does word on the normal goods under normal circumstances.

**(a) What is the main crux in the above passage? [OEB]**

**Ans.** The above passage talks about the exceptions to the law of demand.

**(b) Name the exceptions as shown in the above passage?**

**Ans.** Snob Appeal and Future Expectations.

**(c) Explain the exception as highlighted in the above passage with respect to gold.**

**Ans.** Snob effect refers to the desire to possess a unique commodity having a prestige value. The snob effect is described as a situation where the demand for a certain good by individuals of a higher income level is inversely related to its demand by those of a lower income level. The demand curve will have a positive slope, rather than the typical negatively sloped demand curve of normal goods. This situation is derived by the desire to own unusual, expensive or unique goods. These goods usually have a high economic value, but low practical value. The less of an item available, the higher its snob value. Examples of the snob effect are rare works of art, especially designed sport cars, especially designed clothing made to order, very expensive luxury cars.

**(d) Explain the exception as highlighted during the COVID times.**

**Ans.** If the consumers expect a shortage or scarcity of a particular commodity in the near future, then they would start buying more and more of that commodity in the current period even if their prices are rising. The consumers demand more due to fear of further rise in prices. For example, during emergencies like war, famines, etc., consumers demand goods even at higher prices due to fear of shortage and general insecurity.

**(e) Explain any other exception as not mentioned in the above passage.**

**Ans. (i) Giffen goods:** When price of a giffen goods falls, its quantity demanded also falls.

**(ii) Consumer's ignorance:** The law breaks down when consumers judge quality of the commodity by its price. **(Any one)**

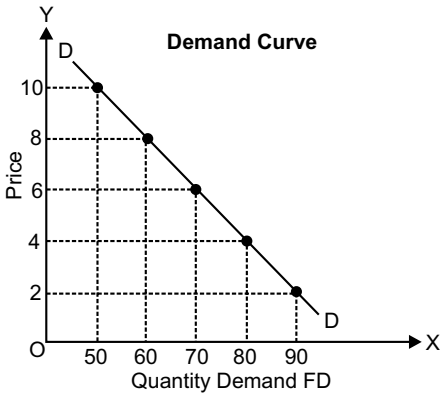
## Solutions for Practice Questions (Topic-1)

### 1 Mark Questions

**2.** Demand can be referred to as that quantity of a commodity which a consumer demands at the given price within a given period of time.

### 2 Marks Questions

**2.** If all other determinants of demand are held constant, then there is an inverse relationship between price of a commodity and its quantity demanded. The slope of the demand curve is negative. This means that with an increase in price of a commodity, its quantity demanded decreases and vice versa.



4. Income Effect refers to change in quantity demanded of a commodity when real income of the consumer changes owing to change in own price of the commodity.

### 3 Marks Questions

2. **(i)** When the income of the consumer increases, the quantity demanded of a normal good also increases. For example, when the consumers income increases from ₹10,000 to ₹15,000, the demand for television will increase.

**(ii)** When the income of the consumer increases, the quantity demanded of inferior good decreases as the consumer shifts their demand to a normal good.

#### Examiner's Comments

- Several candidates did not write the mechanism although they mentioned the effect of change in income on the demand. Some candidates related it with price instead of with income.
- Some candidates related inferior goods with price and not with income. A few candidates related the answer with real income effect which was incorrect.

#### Answering Tip

- The students need to understand that in case of normal goods, the demand increases with increase in income while in case of inferior goods, with the rise in purchasing power the consumer moves to a better substitute, and the demand for the given commodity falls.

5. Income increases, demand for normal goods increases, positive relation or direct relation.
Income increases, demand for inferior goods reduces, inverse or negative relation.
Income increases, demand for necessities remain same. **[ISC Marking Scheme 2015]**

#### Examiner's Comment

- Some candidate defined normal/inferior goods relating with price, though in the question 'income' was clearly mentioned. In some cases, 'price' was measured along the vertical axis to draw the relevant curves of these three types of goods.

#### Answering Tip

- Students must learn the concepts thoroughly to avoid confusion.

## Solutions for Practice Questions (Topic-2)

### 1 Mark Questions

5. Snob Appeal or good with status symbol.

### 2 Marks Questions

2. **(i) Giffen goods:** When price of a giffen goods falls, its quantity demanded also falls.

**(ii) Consumer's ignorance:** The law breaks down when consumers judge quality of the commodity by its price.

### 3 Marks Questions

3. Law of demand can be explained as the relationship between price of the commodity and quantity demanded of a good wherein other factors remain constant. As Price of a commodity rises then demand falls and when price of the commodity falls then demand rises. There is inverse relationship between price and demand for the commodity.

| Price(in ₹) | Demand(in units) |
|---|---|
| 10 | 100 |
| 20 | 80 |
| 30 | 50 |

## Solutions for Practice Questions (Topic-3)

### 1 Mark Questions

2. It is also called as change in quantity demanded. It includes expansion in demand and contraction in demand. Expansion in demand implies demand rises due to fall in price. Contraction in demand implies fall in demand due to rise in price.

### 2 Marks Questions

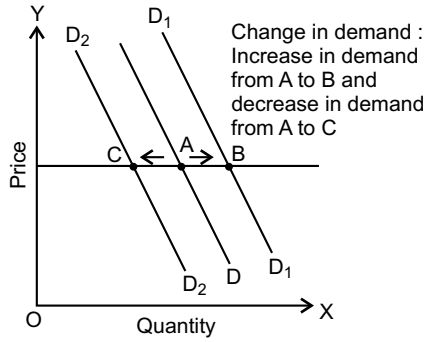
4. Rita was pointing towards the concept of "Increase in demand" because here demand increased due to change in other factors like Covid-19 or due to change in external factors and not due to change in price. And these other factors are responsible for increase in demand.

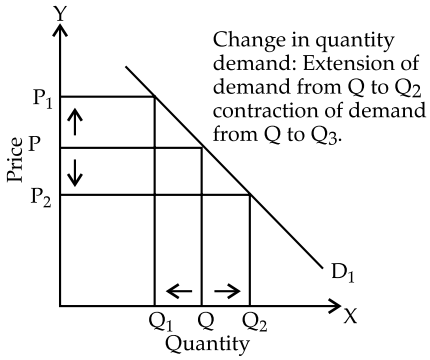
### 3 Marks Questions

3. Change in demand refers to increase or decrease in demand. It is caused by change in determinants of demand other than price of the commodity.

   Change in quantity demanded refers to extension or contraction of demand. It is caused by change in price of the commodity.

   Change in demand implies rightward or leftward shift in demand curve and change in quantity demanded implies movement along the demand curve as shown in following figure:




## Solutions for Objective Type Questions

### [A] Multiple Choice Questions

**2. Option (A) is correct.**

***Explanation:*** Giffen Good is the special type of inferior Good which are very inferior. In such type of goods when the price decreases the demand for such goods also decreases.

### [B] Assertion Reason Questions

**2. Option (C) is correct.**

***Explanation:*** When more of the product is consumed the marginal benefit of the consumer decreases and not rises.

## REFLECTION

**(a)** Do you know the meaning of demand and the factors that affect demand?

**(b)** Can you state the law of demand?

**(c)** Can you explain the various reasons for the downward slope of the demand curve?

**(d)** Do you know the various exceptions to the law of demand?

**(e)** Do you understand the reason for the movement along the demand curve and the shift in the demand curve?